



Chip-based biosensor cuts antibiotic test time

IIT Madras (Chennai, Tamil Nadu) researchers have created a chip-based device, ϵ - μ D, that rapidly tests bacterial resistance to antibiotics in just three hours, far quicker than



Researchers have developed an innovative and affordable microfluidic device that can rapidly determine whether bacteria are resistant or susceptible to antibiotics (Credit: <https://www.iitm.ac.in>)

conventional methods that take up to three days. Built on microfluidic chips with screen-printed carbon electrodes, the system uses electrochemical impedance spectroscopy (EIS) to track bacterial growth in real time. Successfully tested on *E. coli*, *B. subtilis*, and urine samples, the device is now under clinical validation and headed for commercialisation.

Breath test for diabetes detection

Researchers at Penn State (Pennsylvania) and Hebei University of Technology (Tianjin, China) have developed a



This sensor can detect diabetes from a breath sample (Image Credit: Larry Cheng)

laser-induced graphene breath sensor that detects diabetes and prediabetes without the need for blood tests. By capturing acetone in exhaled breath at levels as low as 4ppb, the device can distinguish between patients and healthy individuals within 20 seconds.

Enhanced with zinc oxide and a molecular sieve membrane, it operates effectively in humid conditions, providing a low-cost, non-invasive path to earlier diagnosis and continuous monitoring.

Graphene surpasses traditional semiconductors

Researchers at the National University of Singapore and The University of Manchester (England) have cracked graphene's decades-old mobility barrier, surpassing gallium arsenide (GaAs) semiconductors long considered unbeatable. Using two strategies—twisted bilayer graphene as an electrostatic shield and proximity metallic screening with ultrathin boron

nitride—teams achieved record-breaking electron mobilities up to 60 million cm^2/Vs . The breakthroughs redefine graphene's role in quantum metrology, ultra-sensitive sensing, and energy-efficient electronics, pushing two-dimensional materials into an unprecedented performance era.

Soft electronics redefine brain probes

A new class of brain probes promises to revolutionise neuroscience by capturing neural activity in 3D, rather than in flat planes. Developed through a technique called ROSE (rolling of soft electronics) by researchers at Dartmouth (Hanover, New Hampshire) and partner institutions, the innovation rolls flexible electronics into cylindrical 'Swiss roll' structures. These probes offer hundreds of recording sites with customisable spacing and depth, reducing tissue damage and surpassing rigid 2D arrays—paving the way for deeper neural mapping and next-generation prosthetic applications.

A novel way to record lightning from space

Researchers at Western Sydney University (Penrith, Australia), with support from the US Air Force Research Laboratory, have demonstrated a novel way to record lightning from



Falcon Neuro (located on the right side of this instrument payload on the ISS) can spot individual lightning strikes with high precision (Credit: ASI/ European Space Agency)

space. Their Falcon neuro imager, flown for two years on the International Space Station, uses event-based sensing instead of conventional frame-based cameras. By detecting microsecond changes in brightness, it captures metre-scale lightning footprints and timelines at up to 1000 frames per second, while reducing data loads to just a few Mbit/s.

World's first 6G chip achieves 100Gbps

Chinese researchers from Peking University (Beijing, China) and City University of Hong Kong have developed the world's first 6G 'all-frequency' chip, delivering speeds beyond 100Gbps. Just $11 \times 1.7\text{mm}$ in size, it spans 0.5–115GHz—replacing nine separate systems with a single device. Built on thin-film lithium niobate, it converts wireless signals into optical ones for ultra-fast, stable performance.